Abstracts

Brisbane Powerhouse Opening

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Brisbane has been fortunate in witnessing the opening of an innovative space for "Live Arts". It is backed by Brisbane City Council and is therefore relatively well funded compared to most spaces of this type in Australia.

Its mission is to develop and present bold new work from the Brisbane Community as well as to "import" significant cutting edge work from the rest of Australia (and sometimes the world)

In early May QUT Music staff & students were involved with the opening ritual, which was a multimedia event (additionally featuring Rock n Roll Circus and Arterial Contemporary Public Arts). It was a confluence of youth, community and new work. The process and the work itself will be presented in this talk.

Environmental Infrasound Recordings

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The earth, and indeed the universe, vibrate across a very large frequency range, only a small portion of which is recordable using conventional audio equipment. Nevertheless, very low frequency sound and vibration is routinely recorded for purposes other than our listening pleasure, using other transducers and data formats. In this talk I discuss three sources of very low frequency sound recordings, what was involved in converting them to audible sound, and the results.

One month of seismic data was generously supplied by Arie Verveer, who monitors seismic activity at Kalamunda in Western Australia. As the data format was 16-bit signed integer, only a little manipulation was required to convert this to a soundfile format. The original sample rate of 40 Hz was converted to 44100 kHz. Seismometers have a dipole sensitivity pattern: the North-South data and the East-West data were used for the two audio channels. Within the period of the recording, several large events can be heard, along with a nuclear test, and varying local micro-seismic noise.

Fifty years of barometric data were obtained from the Australian Bureau of Meteorology, measured at Laverton and Williamtown Air Force Bases. The sample period was usually 6 hours. It is doubtful that any antialiasing filtering was applied. The data were supplied in text format, so a simple computer program was written to convert it into a sound file. The recording was sped up so that one year's record corresponded to one second of playback (a factor of approximately 32 million). The atmospheric "tide", the yearly cycle, and sudden pressure changes can all be heard in this short recording.

Tidal spectra, published in the Australian National Tide Tables, were resynthesised (but sped up 32 million times). Of course, spectra are a much cleaner (and less interesting) source of data than waveforms and interest is created in this recording by simulating a circumnavigation of the Australian mainland in a virtual space. Tidal spectra involve more than twenty significant frequency components, the

amplitudes of which vary from port to port. Most tidal components have a period of approximately 12 hours (semi-diurnal) or 24 hours (diurnal). However these components are all “out-of-tune” with each other, creating prominent beats. The south-west of Australia has very weak tides. The north-west has a very strong semi-diurnal tide, with the strongest tide at Derby.

These recordings are not just data transformed into sound. They are reproductions of real acoustic events, occurring on a very large scale in our acoustic environment. In March these recordings were used in a dance performance by Tess de Quincey, entitled “SkyHammer”, at The Performance Space in Sydney.

LaTrobe CD Archive

David Hirst

A selection of recordings form La Trobe Music's collection of tapes have been transferred to CD over the last year. We have tried to put together a mixture of materials on each CD consisting of works from 1976 to 1999, electroacoustic & instrumental/vocal. Most of the works are original pieces by staff and students, and the electronic works constitute about 50% of the works. There are now 10 CDs of material. Our object is to preserve some of the sounds from La Trobe for future listening & research and deposit 4 complete sets in 4 archives/libraries in Sydney, Canberra, & Melbourne.

An Approach to Spatiotemporial Music

Peter McIlwain
School of Music - Conservatorium
Monash University

The artists talk will involve a discussion of the SNet software program which generates music for multispeaker sound systems. The software generates spatiotemporal events (notes) for multispeaker arrays from 8 to 32 speakers. These events are generated by a real time neural network consisting of 32 nodes which develop and evolve spatiotemporal patterns. Notes are generated using a flocking algorithm which determine MIDI values for; pitch, velocity and note duration.

The talk will centre on the following:

- compositional problems in relation spatiotemporal music and the SNet program as one approach to solving these problems.
- a technical overview of the SNet program and its implementation
- an analysis of the behaviour of the neural network
- an analysis of the behaviour of the flocking algorithm
- a demonstration of the program with reference to the "Bird Speaker Pieces" which is submitted for the concert program in the conference.

Programmable Composition

Alistair Riddell

Composition using computers has inherently promoted the idea of a private composition discourse. Since the composer works directly with the computer to produce sound such an idea is often simply taken for granted. It is only when the composer is confronted by a new and explicit language for composition that there arises a concern about whether the language will facilitate the kinds of compositional statements that the composer hopes to make. The primary concern is the language of music rather than the language of musical production.

The use of general computer languages has always been a starting point in any creative endeavour based on the computer but traditionally these languages have been superseded or augmented by more musically orientated versions which in themselves impose a specific compositional paradigm.

Also most electronic composers confront systems which hide or disguise the detail of the musical semantic structure by providing cognitively easier modes of access (graphics or an instrumental interface). Additionally, examples facilitate use of the system by making the logic of the process much more evident. It would be exceedingly frustrating to work out a compositional procedure from the basis of the syntax alone.

So what have the general computer languages in their immediate form got offer the composer? Is there a trend to using these languages in conjunction with special musical functions that is going to result in different musical paradigms?

Distributed Audio Sequencer

Kenny Sabir

The aim of the DASE project was to create a Distributed Audio Sequencer (DASE) that would allow electronic musicians to play together over the Internet. DASE is audio based and is broken into 2 programs. The Castle Application is for playing the audio, while the peasant application is for plotting the beats. These 2 programs can be combined and networked in any configuration. An example of an application of the DASE is for a musician in Sydney to create and edit a drum pattern over the Internet with a musician in London who is creating a bass riff at the same time. The architecture of DASE assumes that the music to be created will be mainly loop based, that is, the song is made up of fixed duration patterns that are often repeated. Therefore, DASE will suit the creation of modern electronic music.

Interactive interface design: Tools and processes to enable the development of alternate input devices

Gavin Sade

This paper will discuss tools and techniques used by practitioners working with interactive media, with a view to developing an understanding of how these tools and techniques can be used to enable the creative process.

The material will be discussed in relation to the field of human computer interface design and focus on the creation of input devices to support the exploration of alternate interaction techniques.

A Survey to the Theory and Practice of Just Intonation

Greg Schiemer
Sydney Conservatorium of Music

This presentation is an introduction to the theory and practice of just intonation showing how it provides a unified theoretical basis for music found not just in the western tradition but in many non-western cultures past and present. It summarises the contribution of just intonation advocates, explains theoretical concepts such as harmonic identity, tuning limits, Partch's tonality diamond, Wilson's combination product sets and Chalmer's tri- and pentatriadic scales, and examines the role of computer technology in the on-going development of just intonation. Musical examples will be limited to the time available.

An Examination of Selected Algorithmic Compositional Processes

Andrew Troedson

This paper will look at several algorithms which have historically been used in the algorithmic composition of music, and will discuss the usefulness of each of these in the creation of music to accompany computer entertainment software. The algorithms will be classified according to their functional basis and it will be shown that each group has particular aesthetic characteristics and therefore are different in their compositional tendencies and potential application.

Musical examples of each of these algorithmic methods will be given to highlight the aesthetic successes and failures of each when assigned to particular musical parameters such as pitch, rhythm, and dynamics.

These observations will then be summarised to suggest a possible direction for the creation of an adaptive algorithmic composition system.

Musical Environments

Richard Vella

Referring to models of perception, *Musical Environments* is a discussion of musicianship which avoids the reliance on traditional music notation or performance skills and discusses music and sound on equal terms. Computer, instrumental, pop, classical music, etc, are examined in terms of sounds in time and space. Part One discusses sounds in space whereas Part Two, using texture as its main thrust, focuses on sounds in time.

Musical Environments is an approach to develop a systematic understanding of musical thinking at the secondary and tertiary level at a time when the Western European art tradition exists on an equal footing with pop, jazz and world music traditions. Referring to a wide range of musical examples including Jimi Hendrix, Beethoven, Australian composers, jazz, pop, computer music, etc, *Musical Environments* offers a generic approach to the discussion of musical structures.

Composition as revelation

David Worrall

"When I look at a starry sky, I love it in a certain way because I know it in a certain way; ... Consequently, I can handle the concepts of things themselves without being in direct possession of them, under the condition that I may conceive of them and feel them from within in some way.... [E]ven if I am incapable of dominating a certain phenomenon, I am capable of obtaining a truth which is inherent to the conceived or observed phenomenon, thanks to a kind of immediate revelation. Henceforth, I can accept and use this, in and as itself." [Iannis Xenakis]

The use of algorithms as formal compositional devices can provide mechanisms for revealing an inner nature: of sound, of material, of process, of our perceptions, our systems of symbols and metaphors. By placing an emphasis on active listening rather than trying to convey/receive messages, they can free an artist from the bounds of mere self-expression; in this sense they are revelatory.